DGLRM

From Bed Rest to Mars:

Development of a Literature-Derived Machine Learning Framework for Predicting Lower-Limb Muscle Atrophy in Spaceflight Analogues

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64. Jahrestagung der DGLRM



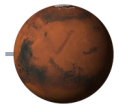
Agenda

Part 1: background and data
Niloufar Ahmadymarzdashty

Part 2: modelling and prediction
Milad Bahari Qaragoz



1 2 3 4 5 6



Intro and motivation
Why predict muscle loss; from bed rest to Mars

Literature research
Range of studies; search and screening

Data extraction and database
Dataset contents; loss per muscle group

ML framework
Testing on unseen campaigns

Models and results
Model comparison; LLM validity checks

Conclusion
Findings, limitations, further work

Motivation: predicting lower-limb muscle atrophy

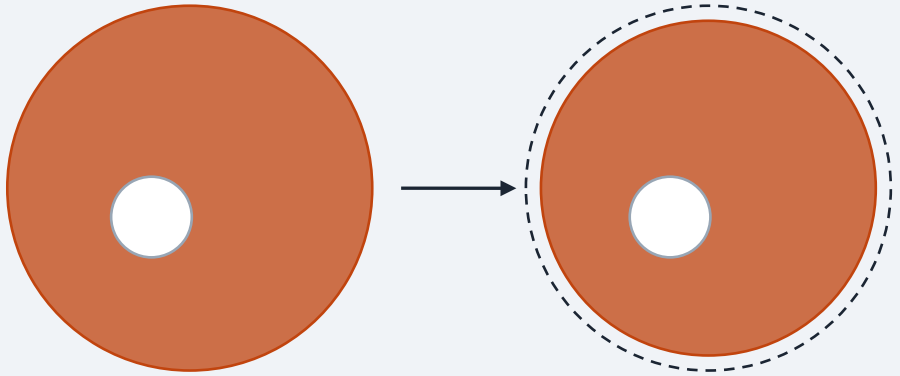
Why prediction matters

- Prevention: identify risk early and leave time to intervene
- Countermeasures: match exercise and nutrition to the predicted risk
- Vulnerable muscles: show which muscles need closer monitoring
- Function: less muscle loss helps preserve strength and power
- Mission planning: estimate muscle loss on long missions with limited exercise equipment

Figure 1. Unloading shrinks leg muscles



Weeks to months in bed; legs carry no body weight



Schematic calf cross-section; dashed line = size before unloading.

From bed rest to Mars: analogue and application

Bed rest: the ground analogue

- Spaceflight data on muscle are scarce
- Head-down bed rest unloads the legs under controlled conditions

Mars: the long-term application

- No early return in an emergency

Figure 2. Moon mission vs Mars mission



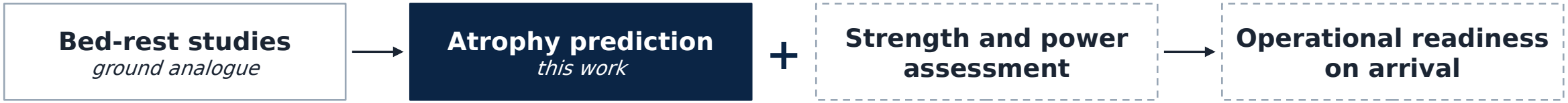


Figure 3. From bed-rest data to crew readiness

Dashed: future use, which needs validation with spaceflight data

Range of unloading durations and studies

Figure 4. Unloading range in the abstract and in the final dataset

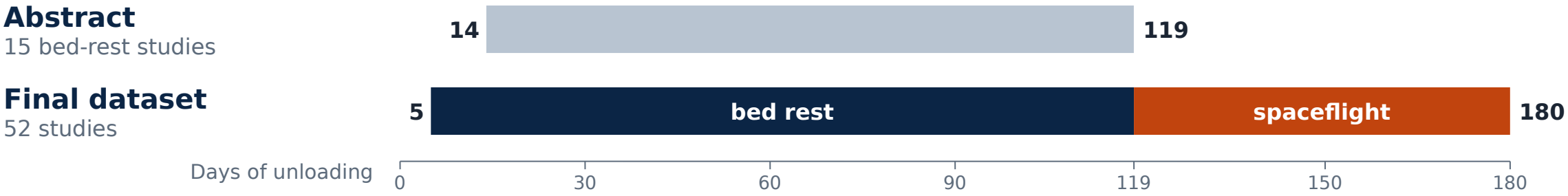
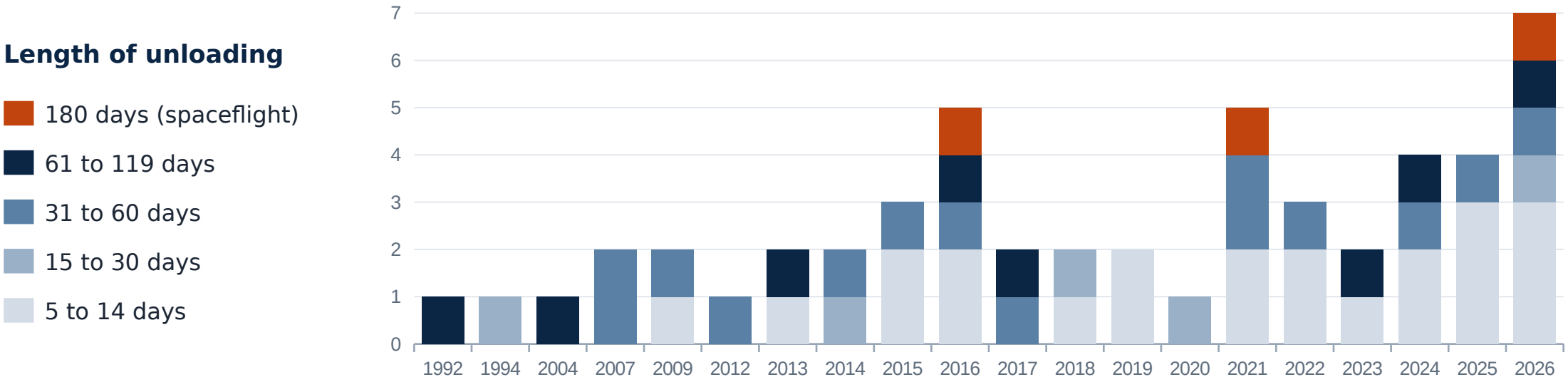


Figure 5. Studies published per year, by length of unloading

Years without studies omitted



Literature search and study selection

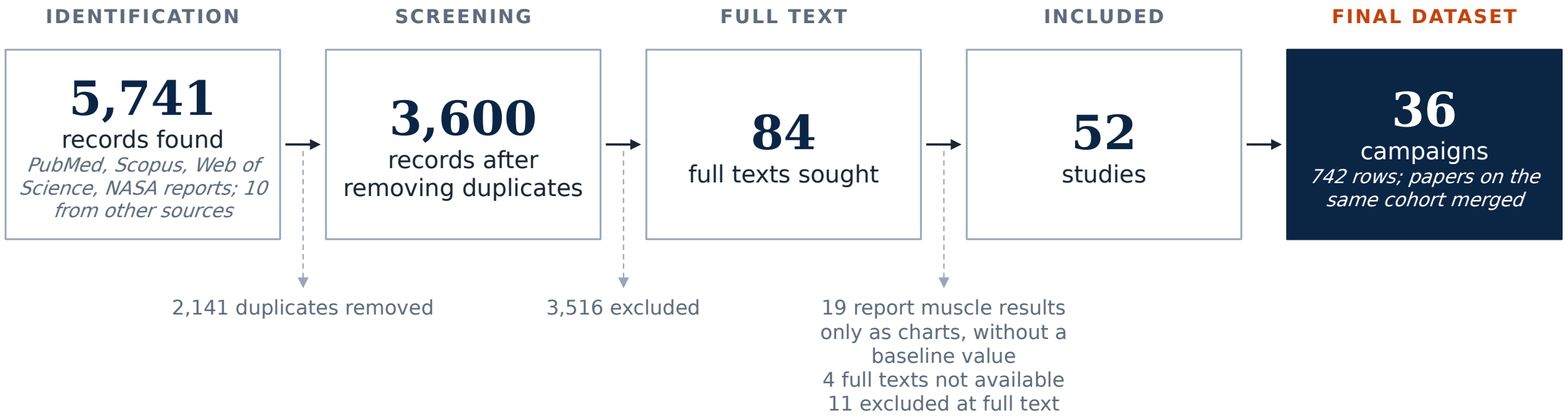


Figure 6. Flow of records, in the style of PRISMA 2020. Full-text counts are not yet reconciled with the dataset.

Dataset composition

Table 1. Final dataset at a glance

Feature	n
Muscles or muscle groups	**51**
Rows: control / countermeasure	**470 / 272**
Rows: during bed rest / recovery	**478 / 264**

Table 2. Imaging method and size measure

Method	n	Size measure	n
MRI	**655**	Volume	**594**
DXA	**37**	Cross-sectional area	**100**
CT	**30**	Lean mass	**37**
Ultrasound	**20**	Thickness	**11**
Total	**742**	**Total**	**742**

742 rows come from 36 campaigns: one campaign gives rows for several muscles, groups and scan days.

Muscle atrophy by muscle group

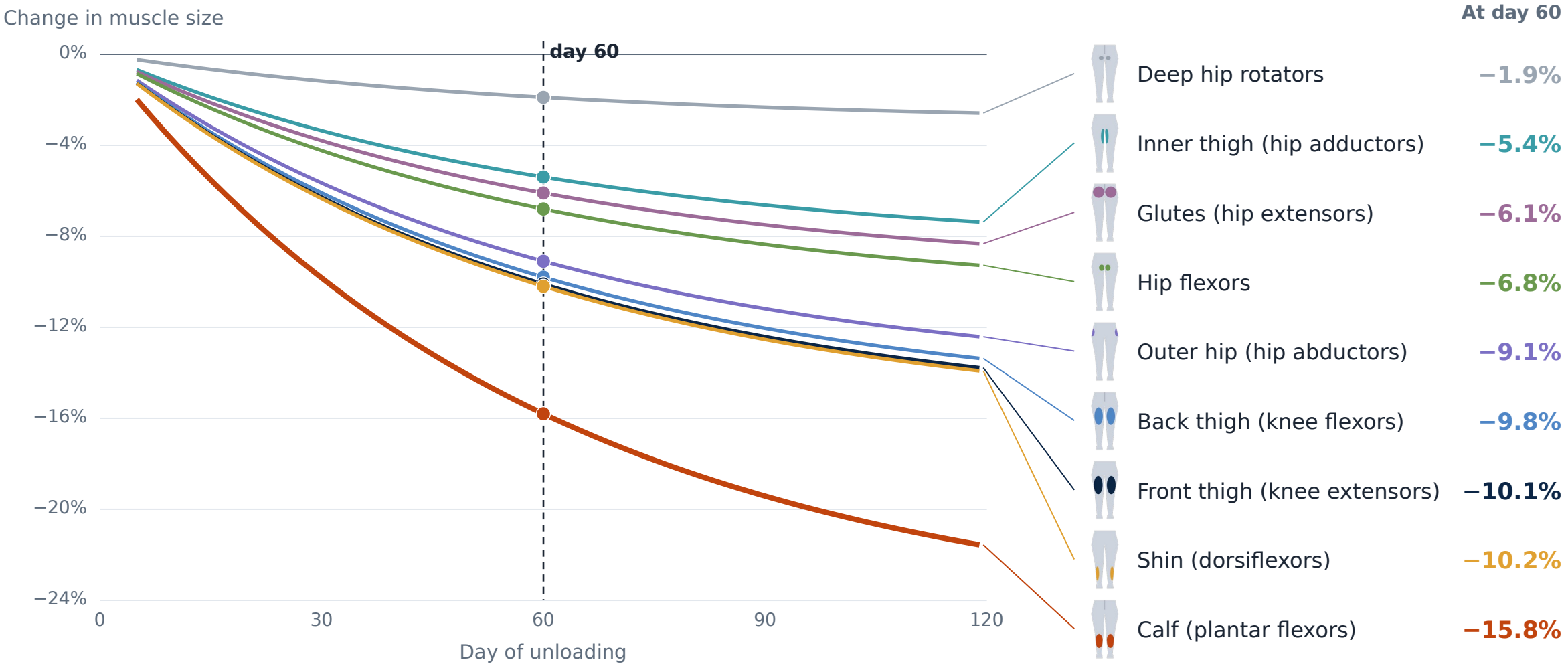


Figure 7. Change in muscle size by muscle group: one shared time course (τ = 60 days) scaled to each group's day-60 estimate; control groups, days 5 to 119.

Evaluation: leave-one-campaign-out validation

Input: one measurement

- Day of bed rest when the scan was taken
- Which muscle group (9, e.g. calf, front thigh)
- Exercise group or no-exercise control
- How it was scanned (MRI, CT, DXA, ultrasound)
- One muscle or several combined

→

Model

- Duration curve: uses the number of days only (the baseline)
- 4 standard ML models
- TabPFN (pretrained network)
- LLM classifier: returns a probability per range

→

Output

How much the muscle has shrunk, in % of its size before bed rest

Average over the group of volunteers, e.g. −10%

Figure 8. Testing: hold out one campaign, train on the rest, repeat 32 times



Grey: 31 training campaigns. Orange: the held-out test campaign.

- Rows of one campaign share volunteers, so whole campaigns are held out. The 346 modelling rows come from 32 of the 36 campaigns.
- Score: average error in percentage points (pp), each campaign counted once. Predicted −10%, measured −13%: error 3 pp.

LLM-based probabilistic classifier: input and output

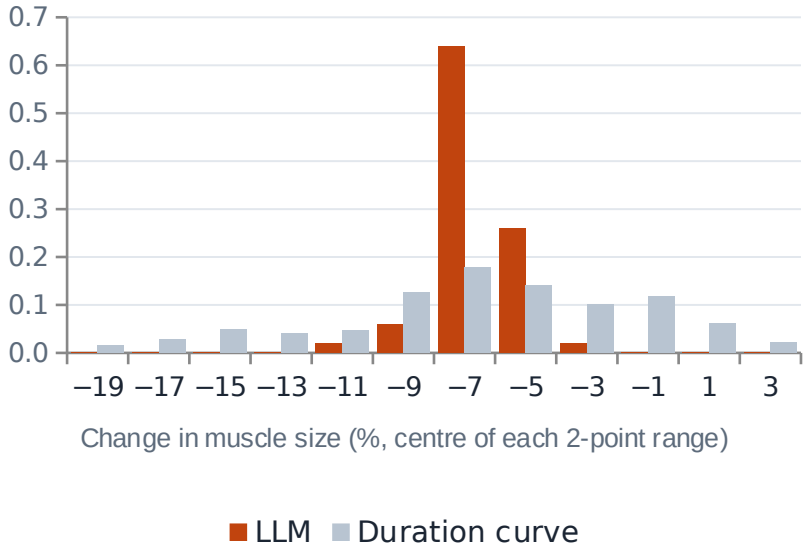
Request (one per measurement)

`participants`
men only, mean age 35, n = 11
`protocol`
head-down tilt bed rest, 21 days, control group
`measurement`
vastus lateralis, MRI cross-sectional area
`target`
day 21 of bed rest
`typical_curve_other_campaigns`
duration curve fitted on the other 31 campaigns
`observations_other_campaigns`
rows from the other 31 campaigns, most similar first, up to about 26,000 tokens
`question`
Which range will the change fall in? 19 options: below −30%, 2-point ranges up to +4%, +4% and above

LLM classifier
fixed version

A classifier: it returns one probability per listed option and never writes text.

Figure 9. Probability per range



- Prediction −6.6%: range midpoints weighted by probability (open ends −31% and +5%)
- Measured −9.4%: error 2.8 pp, the median error

Never sent, checked on every request: the held-out campaign's measurements, and any paper, author or campaign name. All 430 answers are saved, and every result rebuilds from them.

Model comparison on held-out campaigns

Figure 10. Error, pp (lower = better)

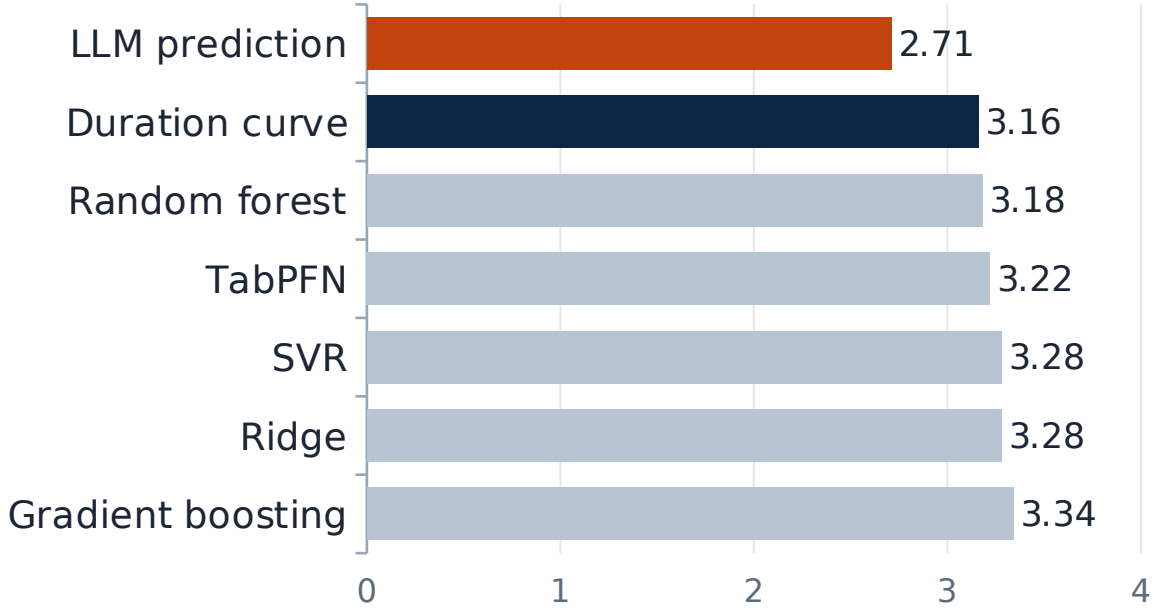
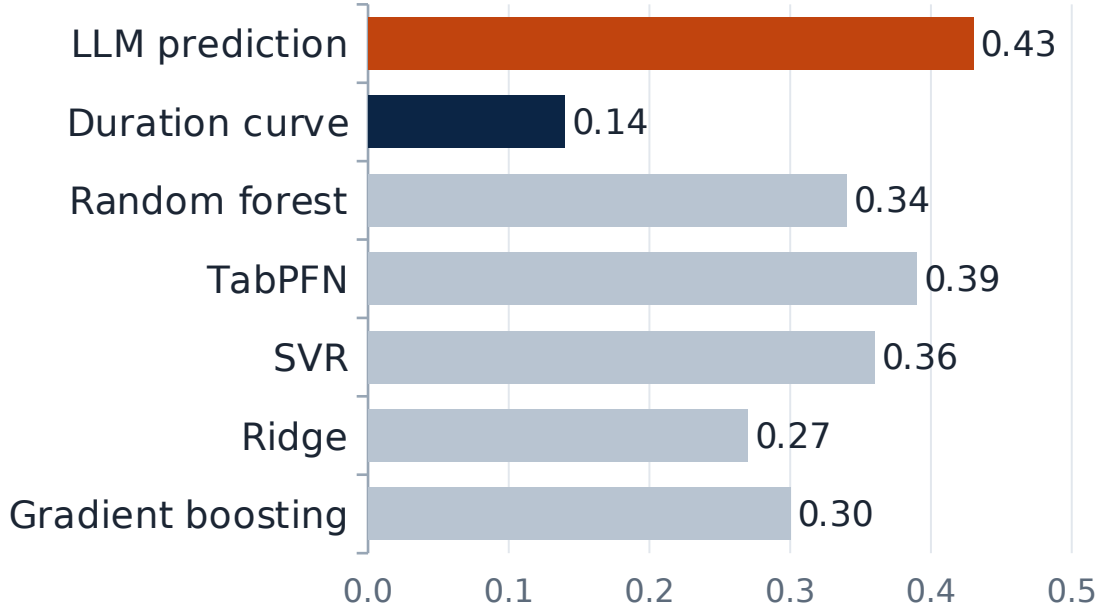


Figure 11. R^2 (higher = better)



- ML models explain more variance (R^2 0.27 to 0.39), but none has lower error than the curve.
- LLM vs curve, scored the same way: 2.71 vs 3.13 pp, so 0.42 pp smaller; closer in 20 of 32 campaigns.
- 95% confidence interval of that difference: 0.13 to 0.73 pp, entirely above zero.

In the paired comparison the curve is scored exactly as the LLM is, which gives 3.13 pp instead of 3.16.

Robustness and validity checks of the LLM prediction

Gain: reduction in error against the curve, in pp (full result 0.42). Each check and its pass rule were fixed before it was run.

Recognition of a known campaign

- Everything identifying a campaign removed: gain still 0.30 pp, interval above zero
- Asked to choose the campaign's name from a list of options: right 19% of the time, where guessing gives 8%

Use of the other campaigns' data

- Their values shuffled: worse than the curve (gain −1.25 pp)
- None of their data given: error 5.85 vs 3.13 pp

Sensitivity to presentation

- Other campaigns' rows in a different order: error changes 3%
- Answer ranges moved by 1 point: error changes 1%

Reproducibility

- 20 identical requests resent: predictions differ by 0.17 pp on average, below the 0.42 pp gain

Limitations of this result

- Added after the ML result was known
- Overconfident ranges: its 80% ranges hold the measured value 62% of the time, so we report only its single-number prediction
- Given a campaign's own earlier scans, it does not use them

Conclusions, limitations and further work

Conclusions

- ML models do not beat a duration-only curve (3.18 vs 3.16 pp)
- The LLM classifier lowers the error to 2.71 pp (curve 3.13 pp)

Limitations

- 32 independent campaigns, mostly young men
- No bed-rest data beyond day 119
- Group averages, not individual astronauts
- Bed rest is not spaceflight

Further work

- Muscle-specific modelling: estimate each muscle's susceptibility and combine it with baseline volumes into total lower-limb volume loss
- Individual and functional prediction: participant-level data over time, linked to strength and function
- Validation on independent campaigns and spaceflight data before any personal risk-assessment tool

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Thank you for listening

Questions are welcome

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Backup: dataset coverage and limitations

Did you miss studies?

- Four sources searched from 2013 on, plus 10 older or open-data studies (sources and queries on the next slide)
- Not covered: Embase (no access)

One campaign dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each campaign one vote; the re-run without it is still to do

Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

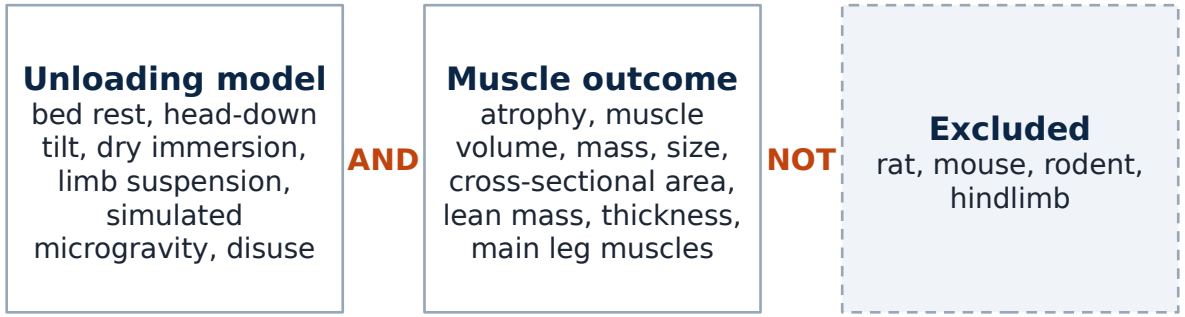
Backup: literature search sources and query

Table 3. Sources (searched 4 September 2026)

Source	Records
PubMed	1,412
Scopus	1,757
Web of Science Core Collection	1,876
NASA Technical Reports Server	686
Earlier studies and NASA open data	10
Total	**5,741**

- Journal databases: publications from 2013 onwards
- Older work enters through studies we held before the search (9) and one NASA open-data campaign
- NASA reports: five short keyword queries, e.g. "bed rest" "muscle volume"
- Not searched: Embase (no institutional access), Cochrane CENTRAL, trial registries, citation chasing

Figure 12. Query: two required blocks



Web of Science string, verbatim

```
TS=(("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR
  OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"
  OR "limb suspension" OR ULLS OR "simulated microgravity"
  OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"
  OR "muscle unloading" OR disuse)
 AND
 (atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"
  OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"
  OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius
  OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```

Backup: dataset expansion and composition

The abstract had 15 studies. Did adding 37 change the analysis after seeing data?

- Eligibility criteria were fixed before screening; the dataset was frozen (14 and 19 September) before the final models
- The curve shape was chosen by a rule declared before fitting

Slide 5 shows 180 days, but the model stops at day 119. Why?

- The 14 spaceflight rows were measured after landing, so they leave the model with the 264 recovery rows
- 8 of the 14 are one back muscle (lumbar multifidus)

Mostly young men: does this apply to women and older crews?

- Rows, not people: men only 567, mixed 113, women only 40 (35 of them WISE-2005), not reported 22
- Healthy young 692, middle-aged 8, older 42: the results describe young men

What exactly is a row's percentage change?

- Follow-up vs the same group's own baseline; negative means loss
- Recomputed from the printed values; where a paper prints the mean of individual changes, that value is kept and labelled
- Rows weighted by number of participants: only 8 of 32 campaigns report the spread of the change

Backup: certainty of the muscle-group curves

Is each line fitted to its own muscle?

- No. The day-60 values are fitted per muscle group (named muscles, 25 campaigns); the lines share one time course (τ = 60 days)
- Separate shapes cannot be estimated: six of nine groups rest on four campaigns or fewer

Why a saturating curve?

- Logarithmic, saturating and spline curves fit equally well (within 1.3 AIC points): the loss slows, but the data cannot say along which curve
- By day 119 the curve reaches 86% of its plateau, −17.3% (95% CI −20.9 to −13.7)

How certain are the day-60 values?

- Calf −15.8% (95% CI −18.9 to −12.7); front thigh −10.1% (−13.6 to −6.6); hip rotators −1.9% (−3.9 to 0.2)
- Calf vs shin, the ankle's opposing pair: 5.6 pp more loss (2.1 to 9.1), p = 0.003
- Hip rotators and outer hip: one campaign each

Is the calf result an artefact of how often it is scanned?

- The order holds after adjusting for duration, countermeasure group and imaging method
- Calf: 78 rows from 13 campaigns, the second best-covered group after the front thigh (86 rows, 22 campaigns)

Backup: significance and relevance of the LLM gain

Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small; we present it as a secondary finding

20 of 32 campaigns is close to a coin flip

- Counting wins only: p = 0.22 (sign test)*
- Using the size of each win: mean gain 0.42 pp, 95% CI 0.13 to 0.73

Does it rest on a few campaigns?

- 3 long MRI campaigns (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI −0.01 to 0.43*
- Only SPRINT is recognised, and with campaign details removed the LLM still wins there (3.79 vs 5.92 pp)

Did it read these papers in training? Was it added after the fact?

- No test can rule out training exposure; hiding campaign details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

** Our calculation from the saved LLM predictions; not in the report.*

Backup: ML results and fairness of the comparison

Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by campaign; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

Why not add age, sex or countermeasure type?

- 32 campaigns leave room for about 3 campaign-level inputs (about 10 campaigns per input)
- Age and sex are fixed within a campaign (271 of the 346 modelling rows men-only), so they act as a campaign label; 100 countermeasure rows spread over 9 types

Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises R^2 but not the error on a new campaign

Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the next control to run

Backup: applicability to Mars mission planning

A Mars transit is about 180 days; the data stop at 119

- Only one campaign reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average